

CLASS X — PRE-BOARD EXAMINATION 2026

SCIENCE (086) — PAPER 1

Date: 03 October 2026 | Time: 3 Hours | Maximum Marks: 80

GENERAL INSTRUCTIONS

1. This question paper contains 39 questions divided into Sections A to E.
2. All questions are compulsory. Internal choices are provided in some questions.
3. Section A: $20 \times 1 = 20$ marks; Section B: $6 \times 2 = 12$ marks; Section C: $7 \times 3 = 21$ marks; Section D: $3 \times 5 = 15$ marks; Section E: $3 \times 4 = 12$ marks.
4. Draw neat, labelled diagrams wherever required.

SECTION A — OBJECTIVE TYPE (20 MARKS)

- Q1.** A student adds dilute HCl to zinc granules. The gas evolved turns a burning splint near bubbles into a pop sound. The gas is: (A) Oxygen (B) Hydrogen (C) Carbon dioxide (D) Chlorine
- Q2.** A convex lens forms an image of an object placed beyond $2F$. Which statement is correct? (A) Virtual, erect (B) Real, inverted and diminished (C) Real, erect (D) Virtual, inverted
- Q3.** Assertion (A): A solution of sodium carbonate is basic in nature. Reason (R): Sodium carbonate reacts with water to produce hydroxide ions. (A) Both A and R are true and R explains A (B) Both true but R does not explain A (C) A true, R false (D) A false, R true
- Q4.** The resistance of a conductor is doubled while potential difference remains constant. Current becomes: (A) doubled (B) halved (C) four times (D) unchanged
- Q5.** Which structure controls the amount of light entering the human eye? (A) Retina (B) Cornea (C) Iris (D) Optic nerve
- Q6.** A solution turns blue litmus red and reacts with sodium hydrogen carbonate producing brisk effervescence. Solution contains: (A) NaOH (B) HCl (C) NaCl (D) $\text{Ca}(\text{OH})_2$
- Q7.** Which of the following represents oxidation? (A) Addition of hydrogen (B) Removal of oxygen (C) Addition of oxygen (D) Addition of electrons
- Q8.** In a magnetic field, a current-carrying conductor experiences maximum force when angle between conductor and magnetic field is: (A) 0° (B) 30° (C) 60° (D) 90°
- Q9.** A plant bends towards a source of light because of the unequal distribution of: (A) Auxin (B) Insulin (C) Thyroxine (D) Adrenaline
- Q10.** Offspring produced by sexual reproduction show greater variation mainly because: (A) only one parent contributes (B) genetic material from two parents combines (C) DNA never copies (D) identical genes
- Q11.** A ray of light passes from glass to air. The ray bends: (A) towards normal (B) away from normal (C) along normal (D) no bending
- Q12.** The image formed on retina is: (A) virtual and erect (B) real and inverted (C) virtual and inverted (D) real and erect

Q13-Q20. [Includes questions on Electrolysis of Sodium, Parallel Resistors Equivalent R, Cerebellum for posture & balance, Testosterone for male secondary traits, Convex Lens Focus, Displacement Reactions, Household parallel wiring, Nephron as functional unit of kidney.]

SECTION B — VERY SHORT ANSWER (12 MARKS)

- Q21.** Why does dry HCl gas not change the colour of dry blue litmus paper, whereas aqueous HCl does?
- Q22.** Two bulbs rated 60 W and 100 W are connected separately to the same voltage. Which bulb has greater resistance? Give reason.
- Q23.** Explain why the sky appears blue during the day.
- Q24.** What is reflex action? Explain the role of the spinal cord in a reflex arc.
- Q25.** Give two differences between asexual and sexual reproduction.
- Q26.** What happens when: (1) Copper is heated in air? (2) The black coating formed is treated with hydrogen?

SECTION C — SHORT ANSWER (21 MARKS)

- Q27.** A student connects a 4 ohm and a 6 ohm resistor: (1) in series, and (2) in parallel. Calculate equivalent resistance in each case.
- Q28.** Draw a labelled ray diagram for an object placed between F1 and 2F1 of a convex lens.
- Q29.** Explain: (1) Why ionic compounds have high melting points? (2) Why do they conduct electricity in molten state?
- Q30.** Explain how the human eye focuses on: (1) a nearby object, and (2) a distant object.
- Q31-Q33.** [Structural isomerism C₂H₆O, Nerve impulse across synapse, 5 ohm resistor energy calculation.]

SECTION D — LONG ANSWER (15 MARKS)

- Q34.** Convex Lens Experiment Analysis & Ray Diagrams OR Working of Human Eye & Myopia Correction.
- Q35.** Iron nail in CuSO₄, ZnSO₄, AgNO₃ solutions: Displacement observations, reactivity order of Fe, Cu, Zn, Ag. OR Metal extraction reactivity series.
- Q36.** Reproduction mechanism in flowering plants from pollination to seed formation with labelled diagram.

SECTION E — CASE / COMPETENCY BASED (12 MARKS)

- Q37.** ELECTRICITY CASE STUDY: V-I graph data analysis and Ohm's law verification.
- Q38.** LIFE PROCESSES CASE STUDY: Carbohydrate digestion, salivary amylase, small intestine absorption, aerobic respiration equation.
- Q39.** MAGNETIC EFFECTS CASE STUDY: Force on current-carrying conductor in magnetic field, Fleming's Left Hand Rule.

PRINCIPAL'S CHALLENGE

Difficulty Level: VERY HIGH. Original practice paper emphasizing application, numericals, diagrams and case studies.